

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Summer - 2024

Course: B.Tech.**Branch : Computer and Allied****Semester : IV****Subject Code & Name: BTBSC404 Probability and Statistics****Max Marks: 60****Date: 20/06/2024****Duration: 3.00 Hr.****Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.**CO 1**

- A) State and prove Multiplication theorem of Probability. 6
- B) An urn contains 8 white and 3 red balls. If two balls are drawn at random, find the probability that 6
- (i) Both are white (ii) both are red (iii) one is of each colour
- C) In a bolt factory, machines A,B,C manufacture respectively 25%, 35% and 40% of the total. Of their total output 5, 4, 2 per cent are known to be defective bolts. A bolt is drawn at random from the product and is found to be defective. What are the probabilities that it was manufacture by 6
- (I) Machine A (II) Machine B or C

Q.2 Solve Any Two of the following.**CO 2**

- A) Find the Mean, Variance and Standard Deviation of the discrete random variable x whose probability distribution is given as 6
- | | | | | | |
|------|-----|-----|-----|-----|-----|
| X | 1 | 2 | 3 | 4 | 5 |
| P(x) | 0.1 | 0.1 | 0.3 | 0.3 | 0.2 |
- B) A car hire firm has two cars which it hires out day by day. The number of demands for a car on each day is distributed as a Poisson variate with mean 1.5. Calculate the proportion of the days on which 6
- (i) Neither Car is used (ii) Some Demand is refused.
- C) The hourly wages of 1,000 workmen are normally distributed around a mean of Rs. 70 and with a standard deviation of Rs.5. Estimate the number of workers whose hourly wages will be: 6
- i) Between Rs. 69 and Rs. 72 ii) more than Rs. 75 iii) Less than Rs. 63
- [(Area between $Z=0$ and $Z=0.4$)=0.1554, (Area between $Z=0$ and $Z=0.2$)=0.0793, (Area between $Z=0$ and $Z=1$)=0.3413 (Area between $Z=0$ and $Z=1.4$)=0.4192]

Q.3 Solve Any Two of the following.**CO 3**

- A)**
- From the following data calculate Karl Pearson's correlation coefficient.

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x	6	2	10	4	8
y	9	11	5	8	7

Also find standard error (S.E) and probable error (P.E).

- B)**
- From the following data calculate rank correlation coefficient.

6

x	32	35	39	60	43	37	43	49	10	20
y	40	30	70	20	30	50	72	60	45	25

- C)**
- Prove that Correlation Coefficient
- r
- is independent of change of origin and scale.

6**Q.4 Solve Any Two of the following.****CO 4**

- A)**
- Obtain the equation of line of Regression of
- y
- on
- x
- and
- x
- on
- y
- for the following data.

6

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

Also find the value of y when $x = 6.2$

- B)** The lines of regression of a bivariate population are $8x - 10y + 66 = 0$ and $40x - 18y = 214$. The variance of x is 9. Find (i) Mean values of x and y (ii) Correlation coefficient between x and y (iii) Standard deviation of x and y .
- C)** At the time of estimation of the regression equations of the two variables x and y , the following results were obtained: $\bar{x} = 90$; $\bar{y} = 70$; $n = 10$; $\sum x^2 = 6360$; $\sum y^2 = 2860$, $\sum xy = 3900$, where x and y are the deviations from the respective means. Obtain the equations.

6**6****Q.5 Solve Any Two of the following.****CO 5**

- A)** In a random sample of 400 persons from a large population, 120 are females. Can it be said that males and females are in the ratio 5:3 in the population? Use 1% level of significance.
- B)** It is claimed that a random sample of 100 tyres with a mean life of 15269 kms is drawn from a population of tyres which has a mean life of 15200 kms and a standard deviation of 1248 kms. Test validity of claim at 5% level of significance.
- C)** In two large populations there are 30% and 25% respectively of fair haired peoples. Is this difference likely to be hidden in sample of 1200 and 900 respectively from the two populations? Use 5% level of significance.

6**6****6*******END*****